

Do Chinese AI companies like DeepSeek require to use 2-4x more power than US firms to achieve similar results to U.S. companies?

Posted by u/Franck_Dernoncourt • 0 points • 15 comments

<https://www.anthropic.com/news/securing-america-s-compute-advantage-anthropic-s-position-on-the-diffusion-rule>:

> DeepSeek Shows Controls Work: Chinese AI companies like DeepSeek openly acknowledge that chip restrictions are their primary constraint, requiring them to use 2-4x more power to achieve similar results to U.S. companies. DeepSeek also likely used frontier chips for training their systems, and export controls will force them into less efficient Chinese chips.

Do Chinese AI companies like DeepSeek require to use 2-4x more power than US firms to achieve similar results to U.S. companies?

Comments

u/Reno772 • 1 points • 2025-05-06 05:05 UTC

I guess so, for now. In the future if Huawei develops and is able to manufacture chips equivalent to what Nvidia has, then probably not

u/butthole_nipple • 1 points • 2025-05-03 09:58 UTC

The also use 1000x more child labor in the process.

u/ThreeKiloZero • 2 points • 2025-05-02 15:41 UTC

The technical answer is yes and no.

They devised tweaks that allowed them to use the technology they could acquire. They also have really top-tier smuggling operations and acquire plenty of modern GPUs. The electricity production in China is cheaper. So while the chips require the same electricity no matter who is using them, the performance they can get from them and the cost per kilowatt-hour, China might actually have an advantage over the rest of the world.

While other countries have been fighting internally over regulations and allowing corporate interference

with public education and health, China has spent decades improving both at a blistering pace. They are now reaping the rewards. That translated into smarter, healthier workers who were able to engineer their way to producing some really great results. That won't stop.

Even though we are talking about AI the main resource people need to be thinking about is still the human.

I think China is going to make a move on Taiwan, not just because they want the country back, but because of the tech stack. If they take it, it's unlimited. chips they need supply of the AI for the future. If they destroy it, they hamstring everyone else. Luckily, the machines that make the worlds' supply of chips possible don't come from Taiwan.

So, IMO, the real race is still on to make the smartest humans and move chip fabs out of China and to their home country the fastest. The electricity and processing bottleneck will be transient. It's gonna spike and then be irrelevant.

u/phovos • 16 points • 2025-05-02 15:14 UTC

Sure but that's all well and fair, China kept advancing their electrical grid and their capabilities while the USA ressted on their laurels and now China is eclipsing us in all elements of energy except perhaps extraction. China has destruyored USA at power generation.

<https://ourworldindata.org/energy/country/china>

https://www.reddit.com/r/Infographics/comments/1j4rpg9/china_vs_usa_total_electricity_generation_by_year/

u/Hadrollo • 6 points • 2025-05-02 15:41 UTC

Yeah, with so much focus on the US in online media, it's easy to overlook that the EU has made the same mistake. I remember as a child hearing about how Germany was heavily investing in its national infrastructure. They did, and they became one of the most competitive nations in the world because of it. Unfortunately, they've been reaping the rewards for so long that many have forgotten the investments they made in the first place.

It **does** look like change is on the horizon, and they're renewing domestic infrastructure as a part of their reinvestment in domestic defence, but they're a long way off China's level.

u/phovos • 1 points • 2025-05-02 15:45 UTC

I hope Europe can change its ways because if it can't then my country the USA is out of luck.

****pokes germany witha stick****

You know, Germany, if you were really concerned about your domestic/national energy security you could have half-finished a half-dozen nuclear reactors in the time since the NordStream was sabotaged.

u/PermanentLiminality • 16 points • 2025-05-02 15:03 UTC

Electricity rates in China are low. It's not really an issue. I pay 40 cents per kwh and in China it looks to be more like 8 cents.

u/holbthephone • 1 points • 2025-05-03 21:24 UTC

Your point is valid but make sure you're only comparing industrial electricity rates, not residential. Residential rates can be 2-4x higher than industrial in the US

u/ProsodySpeaks • 3 points • 2025-05-02 14:52 UTC

I don't know about deployment and use, but I thought training deepseek was orders of magnitude cheaper than openai etc?

u/Comfortable-Mine3904 • 43 points • 2025-05-02 14:26 UTC

yeah measured in wattage this seems like a reasonable estimate. That said, wattage isn't really a constraint so it not a bottleneck in the process.

Yeah your car took 2x the gas compared to a hybrid to drive the same distance. But if the hybrid cost 10k more your all in cost might still be lower

u/holbthephone • 1 points • 2025-05-03 21:23 UTC

Don't forget the different "gas prices" - electricity is far cheaper in China than here, because they don't have as much red tape to get new power generation online

u/DigThatData • 3 points • 2025-05-02 18:36 UTC

> But if the hybrid cost 10k more your all in cost might still be lower

but also pays for itself after traveling some distance. Your "all-in" costs are only lower if this is the only trip you have purchased the car for, which it isn't.

u/fake-bird-123 • -7 points • 2025-05-02 14:03 UTC

I've never used DeepSeek because of its ties to the CCP, but I'd have doubts about believing Anthropic here. Anthropic is getting in with the Trump regime and is pushing similar verbiage that the Trump regime does in this article. I'd need to see much more conclusive evidence than a one-off article of a competitor bashing another competitor, especially when both have polar opposite political affiliations.

u/hinsonan • 13 points • 2025-05-02 14:11 UTC

The model is open so you could use it on an American hosted platform but let's be real OpenAI and Anthropic are just as bad

u/fake-bird-123 • -7 points • 2025-05-02 14:13 UTC

Is it hard for this sub to stay on topic or is it just the two of you that responded to me?

u/hinsonan • 5 points • 2025-05-02 14:16 UTC

If you want me to be smart I'll just tell you to go read the deepseek paper and answer your energy question for yourself

u/absurdherowaw • 12 points • 2025-05-02 14:07 UTC

Also, if you do not use DeepSeek for ties with the CCP in China, then you should also avoid OpenAI/Anthropic/Google for close ties with the current Trump administration in America, simple as.

u/fake-bird-123 • -15 points • 2025-05-02 14:08 UTC

Thanks for the irrelevant comment.

u/Far-Sir1362 • 8 points • 2025-05-02 14:19 UTC

It's only as irrelevant as the one they replied to